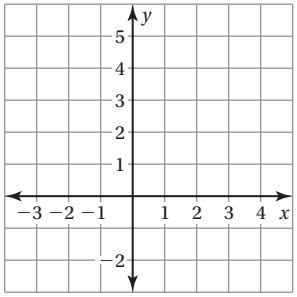
Name _____ Date _____

Lesson Extra Practice:
Polygons in the Coordinate Plane

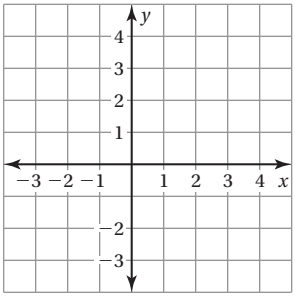
8.6

Draw the polygon with the given vertices in the coordinate plane.

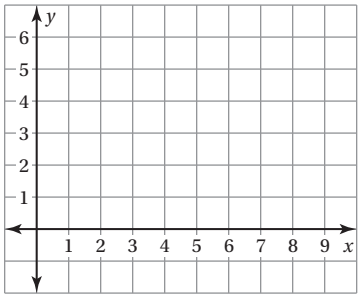
1. $A(-2, 5)$, $B(3, 2)$, $C(0, 0)$



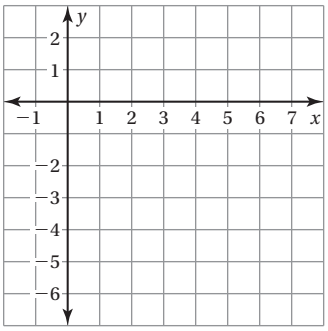
2. $D(3, -1)$, $E(6, 2)$, $F\left(2, \frac{1}{2}\right)$



3. $G(4, 1)$, $H(4, 3)$, $J(9, 3)$, $K(9, 1)$



4. $L\left(4, 2\frac{1}{2}\right)$, $M(4, -6)$, $N(7, -6)$, $P\left(7, 2\frac{1}{2}\right)$



Find the distance between the points.

5. $F(1, 0)$, $G(6, 0)$

6. $J(-3, 1)$, $K(-3, -3)$

7. Your friend finds the distance between $(-2, 4)$ and $(3, 4)$. Is your friend correct? Explain your reasoning.

$|-2| + |3| = 2 + 3 = 5$
The distance between $(-2, 4)$ and $(3, 4)$ is 5.

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Find the perimeter and the area of the polygon with the given vertices.

8. $E(0, 0)$, $F(7, 0)$, $G(7, 2)$, $H(0, 2)$

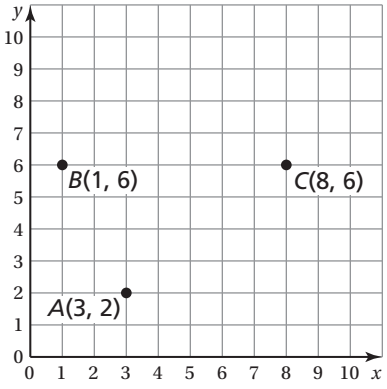
9. $P(-4, 5)$, $Q(-4, -9)$, $R(8, -9)$, $S(8, 5)$

10. You design a rectangular courtyard using a coordinate plane. You plot three of the vertices of the courtyard at $F(1, 0)$, $G(5, 8)$, and $H(1, 8)$. The coordinates are measured in yards.

a. Find the fourth vertex.

b. What is the perimeter and the area of the courtyard?

11. The coordinate plane shows three vertices of a parallelogram. Find two possible points that could represent the fourth vertex.



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